# Workshop 2 – Day 3

## The Merged Model Framework
## Applications and Disaggregation





UNIVERSITY OF
COPENHAGEN

Jonathan Wenzel Pedersen

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# Plan for today

1. **Complete the setup guide**
   Work through the full guide and record when you finish.
2. **Discuss your application ideas**
   What can the model answer, and what would require an extension?
3. **Repeat the disaggregation walkthrough**
   Split agricultural GDP into rice and non-rice agriculture.
4. **Practise disaggregation**
   Apply the same method to agricultural exports.
5. **Review the workshop**
   Share your feedback and remaining questions.
6. **Look ahead to March**
   Discuss the next steps.

# Complete the full setup guide

## Work through the guide from A to Z as quickly as possible.

Open the setup guide (PDF)

The guide is also available in the workshop materials.

1. Wait for the common start signal.
2. Complete every step in the guide, including running the model and setting up the scenario.
3. Finish by opening the **scenario report in your browser**.

**Complete every step**

Follow the full guide even if you have done it before. Ask for help as soon as you get stuck.

# Submit only when you have finished

**WAIT until the full guide is complete**

**Submit only after completing every step and opening the scenario report.**

**When you have finished**

1. Scan the QR code or click the button.
2. Enter your name.
3. Click **Submit** once.

I have completed the full guide

Setup guide completion form

**Submit immediately when finished. An early submission records an incorrect finishing time.**

# Your Ideas and the Model

# Your idea: What would 10–12% GDP growth require?

**What investment and financing would be needed to support a higher GDP growth target?**

**What we can examine**

Change the assumed GDP growth path and examine the implied investment requirements, public and private financing, domestic credit, and foreign debt.

**How to interpret the result**

- GDP growth is an input to this experiment.
- The model calculates the associated requirements and financing implications.
- We must then assess whether those requirements are economically plausible.

# Your idea: Can agricultural investment raise GDP?

**What happens to agricultural GDP if agricultural investment or FDI increases?**

**A constraint of the current model**

Sector GDP growth follows an assumed path. Increasing investment therefore does not automatically generate higher GDP growth.

**What would we need to add?**

- A relationship connecting investment to capital accumulation and production.
- Assumptions about how productive the additional capital would be.
- For FDI, a link between the financing inflow and investment in the relevant sector.

# Your idea: Could better productivity reduce investment needs?

**Could Vietnam support the same GDP growth with less investment?**

**What we can examine**

Change the capital productivity assumption and compare the investment and financing required for the same GDP growth path.

**How to interpret the result**

- Better capital productivity means less investment is required for a given increase in output.
- The model can calculate the resulting financing implications.
- The size of any productivity improvement from AI or technology must be justified using external evidence.

# Your idea: Climate policy and green growth

**How would low-carbon rice, resource constraints, or a greener economy affect growth and investment?**

**Beyond the current model's intended scope**

The current Merged Model does not represent the emissions, resource constraints, and production choices needed for these climate-policy questions.

Splitting agriculture into rice and non-rice agriculture gives us more detail. Climate analysis also needs equations linking production, energy, resources, and emissions.

**Looking ahead**

Next year's CGE model will provide a framework for examining these interactions, with the relevant environmental data and mechanisms.

# Which questions can the Merged Model answer?

**Well suited**

Given a GDP growth path, what investment, savings, borrowing, and credit would be required?

**Can examine**

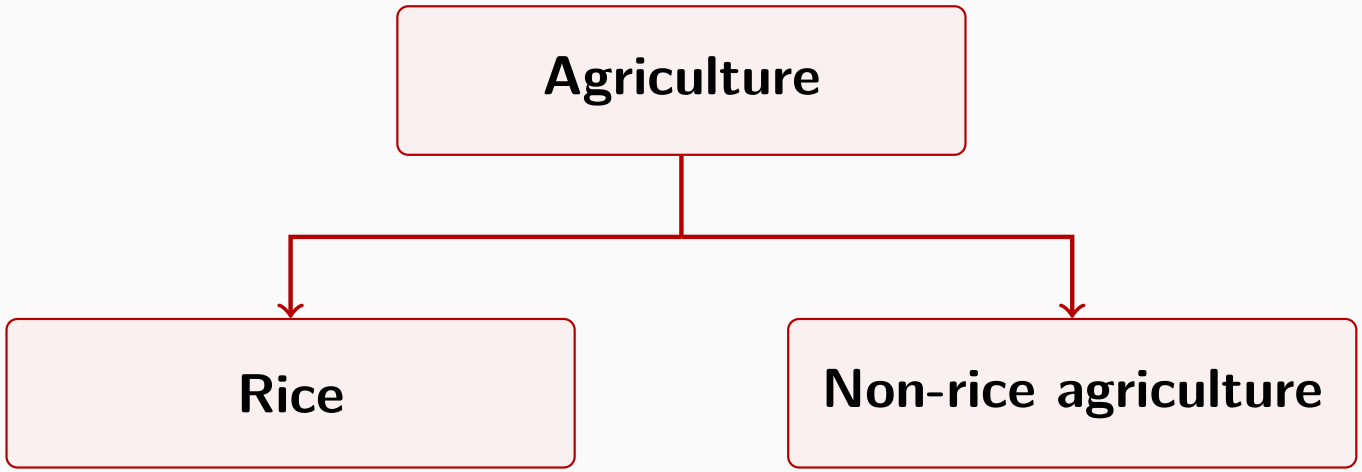
Changes in existing assumptions

**Require additional mechanisms or another model**

Sources of growth and environmental effects

# Disaggregation Walkthrough

# Splitting agricultural GDP




The two groups add up to agricultural GDP.

# Check the split in two runs

**Run 1: Reproduce the original path**

Give both components the original agricultural growth rate in every year. Does agricultural GDP match the original run throughout the projection?

**Run 2: Change rice growth**

Lower rice value-added growth in 2027. Before running, predict how much agricultural GDP growth will change. What determines the size of the effect?

# Disaggregation Exercise

# Split agricultural exports

**Repeat the GDP walkthrough for exports:** rice and non-rice agricultural exports.

1. **Excel:** add rice and non-rice export rows (30% / 70% of agricultural exports).
2. **`sets.jl`:** split `:agriculture` in `sectors`.
3. **`reader.jl`:** map both groups in `SECTOR_EXPORT_SERIES`.
4. **`assumptions.jl`:** give both the original agricultural export growth.
5. **Run:** do rice + non-rice and total exports match the original run?

**Once your paths match**

Lower rice export growth in 2027 in `scenario.jl`. Predict the effect on agricultural and total exports first.

# Workshop Review

# Your feedback and remaining questions

## Please complete the final review

- How confident do you feel using and extending the model?
- What questions or difficulties remain?
- What worked well, and what could we improve?



Scan to open the review

Open the final review

**This is not a test.** Names and email addresses are not collected.

# After this workshop

- **Workshop 2 is complete**
  You have practised running, changing, and disaggregating the Merged Model.
- **Need to add new equations?**
  Contact me and we can arrange a Zoom or Teams meeting to work through your assignment. (jwp@econ.ku.dk)
- **Next week**
  ESAM and static CGE modelling with Dirk.
- **Back in March**
  Introducing the dynamic CGE model **GreenREFORM-VN**.



# Thank you for today